

JEE MAIN REVISION MASTER: HEAT AND THERMODYNAMICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	First Law & Thermodynamic Processes	14 PYQs	First Law
High	Kinetic Theory of Gases & Thermal Radiation	11 PYQs	KTG & Radiation
Moderate	Carnot Engine & Second Law Efficiency	3 PYQs	Heat Engines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
First Law & Thermodynamic Processes	$\Delta Q = \Delta U + W$, $\Delta U = n C_v \Delta T$, $W_{\text{iso}} = nRT \ln \left(\frac{V_2}{V_1} \right)$, $W_{\text{adi}} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$
Kinetic Theory of Gases & Thermal Radiation	$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$, $v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$, $\lambda_m T = b$, $E = e \sigma A T^4$, $\frac{T_1 - T_2}{t} = K \left(\frac{T_1 + T_2}{2} - T_0 \right)$
Carnot Engine & Second Law Efficiency	$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$, $\text{COP} = \frac{1}{\eta}$, $\frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$, $\beta = \frac{1}{\eta}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 First Law & Thermodynamic Processes (High Yield — 14 PYQs)

Core Principle: First law of thermodynamics, molar heat capacities ($C_p - C_v = R$), and work done in cyclic/open processes.

■ **Caution / JEE Pitfall:** Internal energy $\Delta U = nC_v\Delta T$ depends ONLY on temperature change ΔT , regardless of whether the path is isobaric, isothermal, or adiabatic.

■ **Speed Trick:** In cyclic processes on a P-V diagram: clockwise cycles give positive net work (heat engine); counter-clockwise cycles give negative work (refrigerator).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In thermodynamic processes, which of the following statements is not true?..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "A gas is taken from state A to state B along two different paths 1 and 2. The heat absorbed and work done by the system along these two paths..." [Ans: N/A]

3.2 Kinetic Theory of Gases & Thermal Radiation (High Yield — 11 PYQs)

Core Principle: Molecular speeds, energy equipartition $E = (f/2)nRT$, Stefan-Boltzmann law, Wien's displacement law, and Newton's law of cooling.

■ **Caution / JEE Pitfall:** Degrees of freedom: Monatomic $f = 3$ ($\gamma = 5/3$); Diatomic at room temp $f = 5$ ($\gamma = 7/5$); Polyatomic non-linear $f = 6$ ($\gamma = 4/3$).

■ **Speed Trick:** Net heat radiated by a body at temperature T in surroundings T_0 is $P_{\text{(net)}} = e\sigma A(T^4 - T_0^4)$.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "The graph of pressure P and volume V of 1 mole of an ideal gas at constant temperature is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "At 27°C temperature, the mean kinetic energy of the atoms of an ideal gas is $E_{(1)}$. If the temperature is increased to 327°C ..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Three metal rods of the same material and identical in all respects are joined as shown in the figure. The temperatures at the ends of these..." [Ans: N/A]

3.3 Carnot Engine & Second Law Efficiency (Moderate Yield — 3 PYQs)

Core Principle: Carnot cycle efficiency limit, reversible refrigerators, and Clausius/Kelvin-Planck statements.

■ **Caution / JEE Pitfall:** Temperatures T_1 (source) and T_2 (sink) MUST strictly be substituted in Kelvin (K), never in Celsius ($^\circ\text{C}$).

■ **Speed Trick:** To maximize Carnot efficiency, lowering sink temperature T_2 by ΔT is more effective than raising source temperature T_1 by the same ΔT .

Representative Source PYQs:

- [2021 Shift] Q#4 (MCQ): "A number of Carnot engines are operated at identical cold reservoir temperatures (T_L). However, their hot reservoir temperatures are $k \dots$ " [Ans: N/A]
- [2018 Shift] Q#4 (MCQ): "A Carnot engine takes 300 calories of heat from a source at 500 K and rejects 150 calories of heat to the sink. The temperature of the sink ..." [Ans: N/A]
- [2017 Shift] Q#3 (MCQ): "For which combination of working temperatures, the efficiency of Carnot's engine is the least?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and air ignited by a spark III. The leaking of gas from the kitchen cylinder. IV. The transfer of heat from one heated part of a liquid to the other colder part. Which amongst these processes are irreversible in nature?

- (A) I and II
- (B) I and III
- (C) II, III and IV
- (D) I, II, III and IV

Correct Answer: None

Solution Analysis: All the given processes are irreversible in nature. Free expansion of a gas is irreversible because the gas spreads into the available space on its own. It cannot return to its original volume without some external work being done. Combustion of petrol and air is also irreversible. Once petrol burns, it forms new substances like carbon dioxide, water vapour, etc. These products cannot change back into petrol and air by simply reversing the process. Leaking of gas from a kitchen cylinder is irreversible because the gas moves from high pressure inside the cylinder to lower pressure outside. It w

Benchmark Problem 2: KCET 2026 (Q#3)

The graph of pressure P and volume V of 1 mole of an ideal gas at constant temperature is

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: According to the ideal gas equation, $PV=nRT$. Here, the gas is 1 mole, so $n=1$. The temperature is constant, so T is also constant. Since R is already a constant, the whole quantity nRT becomes constant. So, we get $PV=\text{constant}$. This means that when volume V increases, pressure P decreases, and when volume V decreases, pressure P increases. So, pressure is inversely proportional to volume: $P \propto (1/V)$. The graph of P versus V for this relation is a rectangular hyperbola. Among the given graphs, Graph-IV shows this rectangular hyperbola.

Benchmark Problem 3: KCET 2021 (Q#4)

A number of Carnot engines are operated at identical cold reservoir temperatures ($T_{(L)}$). However, their hot reservoir temperatures are kept different. A graph of the efficiency of the engines versus hot reservoir temperature ($T_{(H)}$) is plotted. The correct graphical representation is

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: Efficiency of a Carnot's engine is given as $\eta = 1 - \frac{T_{(L)}}{T_{(H)}}$ where, $T_{(L)}$ = temperature of cold reservoir and $T_{(H)}$ = temperature of hot reservoir. At constant $T_{(L)}$, we can say that, with the increase in the value of $T_{(H)}$, η will also increase. So, the graph between efficiency and $T_{(H)}$ is correctly depicted in option (b).